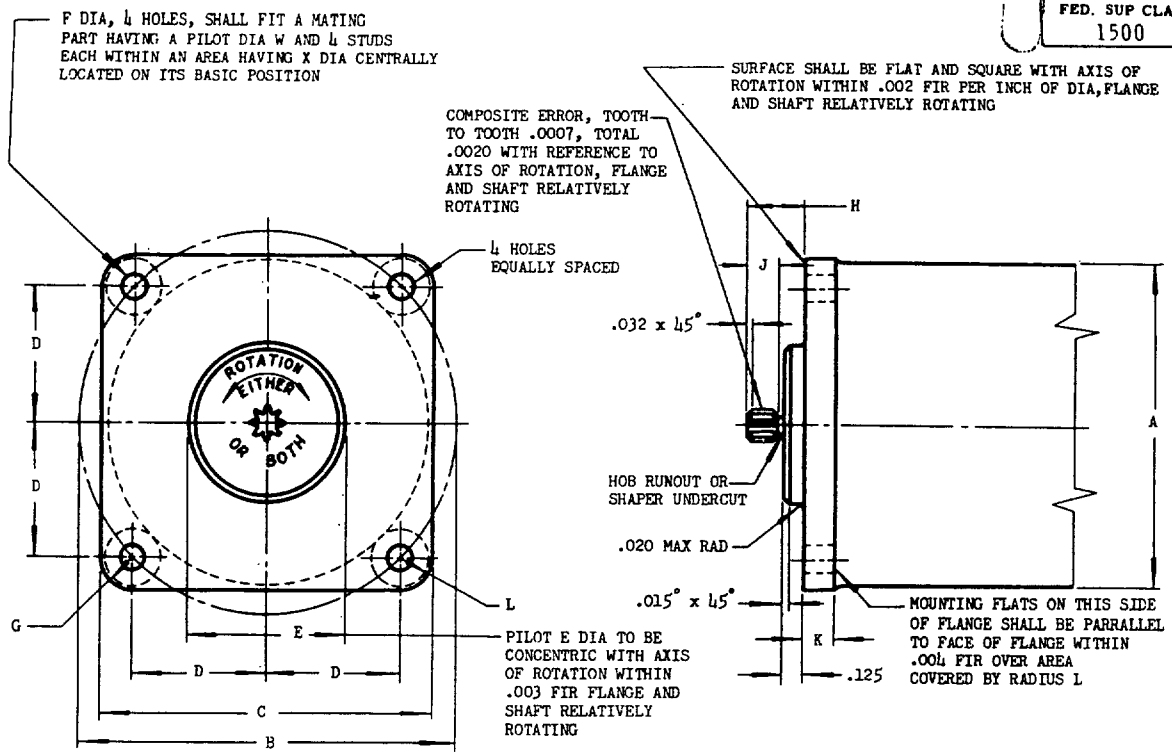
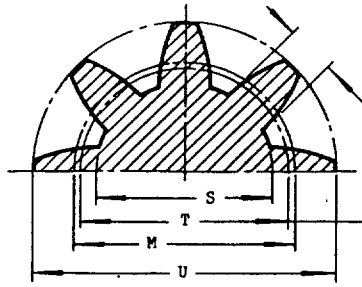
FED. SUP CLASS.
1500



SPLINE-PINION DATA

N = NUMBER OF TEETH
P = DIAMETRAL PITCH
20° PRESSURE ANGLE

ROCKWELL C OR EQUIVALENT HARDNESS:
SURFACE 45 MINIMUM 55 MAXIMUM
CORE 55 MAXIMUM



SPLINE-PINION DETAIL

(a) A IS GREATEST DIAMETER OR DIMENSION OF ACCESSORY EXCLUSIVE OF ATTACHED DEVICES.
(b) OUTSIDE DIAMETERS OF SMALL PINIONS ARE ENLARGED TO AVOID UNDERCUT, TO MAINTAIN STANDARD CENTER DISTANCE, THE GEAR DIAMETERS MUST BE DECREASED BY THE AMOUNT OF THE PINION ENLARGEMENT GIVEN AT Q. IF MATING GEARS ARE MADE WITH STANDARD TOOTH PROPORTIONS, THE CENTER DISTANCES MUST BE INCREASED APPROXIMATELY BY 1/2 Q.
(c) OTHER MASTER GEARS MAY BE USED IF CENTER DISTANCE IS PROPERLY RECALCULATED.
(d) R MAX HAS BEEN REDUCED FROM THEORETICAL BY TOTAL COMPOSITE ERROR OF PINION ONLY, RANGE RESULTS FROM SIZE VAR ONLY.
(e) RANGE OF CENTER DISTANCE IS THAT RESULTING FROM RMS SUMMATION OF ALL VARIABLES IN THIS STRUCTURE.

SIZE DETERMINATION OF MATING GEAR MUST RECOGNIZE FURTHER VARIATIONS IN ITS SUPPORTING STRUCTURE.

SPLINE-PINION DIMENSTIONS ARE BASIC MAXIMUM.

PINION DIMENSIONS ARE ADEQUATE FOR TYPE SHOWN BUT ALL PINIONS SHALL BE CHECKED FOR APPLICATION.

REMOVE ALL BURRS AND SHARP EDGES.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±2°.

NOMINAL USE OF THIS DRIVE FOR AC OR DC ELECTRICAL MOTORS.

(A) INACTIVE 1 JAN 1977 FOR NEW DESIGN SEE SAE STANDARDS AS959 THRU AS972, AS976 AND AS977. (CONTINUED APPLICATION OF THIS STANDARD ON CONTRACTS IN EFFECT PRIOR TO 1 JAN 1977 IS PERMISSIBLE).

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THIS IS A DESIGN STANDARD. NOT USED AS A PART NUMBER

APPROVED 30 Nov 55 REVISED (A) 26 NOV 1976

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS33568 (ASG)
Navy -AS Air Force 99	A - N - AF -	DRIVE, SQUARE MOUNTING FLANGE WITH INVOLUTE SPLINE-PINION	
PROCUREMENT SPECIFICATION NONE		SUPERSEDES: AND10457	SHEET 1 OF 2

DD FORM 672-1
1 OCT 52

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1500

TYPE	(a) A MAX DIA	B REF DIA	C REF	D BASIC	E +.000 -.002 DIA	F +.015 -.005 DIA	G +.000 -.030 RAD	H ±.030	J MIN	K MAX	K MIN	L MIN RAD	M PD THEO	N	P	(b) Q THEO
I	2.032	2.406	2.000	.850	.999	.161	.150	.438	.188	.172	.086	.190	.1111	8	72	.0148
II	2.532	3.000	2.500	1.060	1.499	.191	.187	.468	.219			.200	.1667		48	.0222
III	3.170	4.000	3.578	1.414	2.874	.344	.375	.500	.250	.370	.185	.400	.2500	12	40	.0124
IV	4.045	5.000	4.286	1.767	3.249			.625	.313				.3250	13		.0122

TYPE	(d) R MAX	(d) R MIN	S DIA THEO	T MAX DIA	U DIA MAX	U DIA MIN	W ±.0001 DIA	X ±.0001 DIA	MEAS WIRE DIA	MEAS OVER 2 WIRES MAX	MEAS OVER 2 WIRES MIN	MASTER GEAR (c) NO. TEETH	MASTER GEAR (c) TOOTH THKNS	(e) CENTER DISTANCE WITH MASTER THEO	(e) CENTER DISTANCE WITH MASTER MAX	(e) CENTER DISTANCE WITH MASTER MIN
I	.0266	.0263	.0885	.1056	.1535	.1498	.9995	.1522	.02667	.1591	.1585	108	.0216	.8127	.8134	.8054
II	.0402	.0399	.1348	.1580	.2304	.2266	1.4995	.1791	.0400	.2390	.2384	96	.0327	1.0940	1.0948	1.0866
III	.0367	.0363	.2084	.2358	.3040	.3002	2.8745	.3310		.3167	.3160			1.1311	1.1346	1.1261
IV	.0431	.0427	.2730	.3063	.3868	.3831	3.2495		.0480	.4035	.4029	80	.0393	1.1681	1.1697	1.1593

(A) SEE SHEET 1 FOR CHANGES

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